Paper Review and Code Reproduction

End-to-End Safe Reinforcement Learning through Barrier Functions for Safety-Critical Continuous Control Tasks

Original paper by

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AAAI 2019 | arXiv:1903.08792v1

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Motivation

1. Model-free reinforcement learning improves long-term return through exploration, but training actions can violate safety constraints before an effective policy is learned.
2. Control barrier functions (CBFs) enforce safety constraints[1], but depend on a dynamics model and bounds on model error.
3. Retaining past CBF corrections in later policy updates allows RL to improve the previously deployed safe controller.

Objective

Improve control performance through reinforcement learning while satisfying safety constraints during training under model uncertainty.

[1] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, "Control Barrier Function Based Quadratic Programs for Safety Critical Systems," *IEEE Transactions on Automatic Control*, vol. 62, no. 8, pp. 3861–3876, 2017.

Problem formulation

Consider discrete-time, control-affine dynamics

$$s_{t+1} = f(s_t) + g(s_t)a_t + d(s_t), \tag{1}$$

where f, g define the nominal dynamics and $d(s_t)$ is the unknown state-dependent model residual.

Performance

$$\max_{\pi} J(\pi) = \mathbb{E}_{\tau\sim\pi}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)\right].$$

Learn a policy that maximizes expected return from sampled trajectories.

Safety

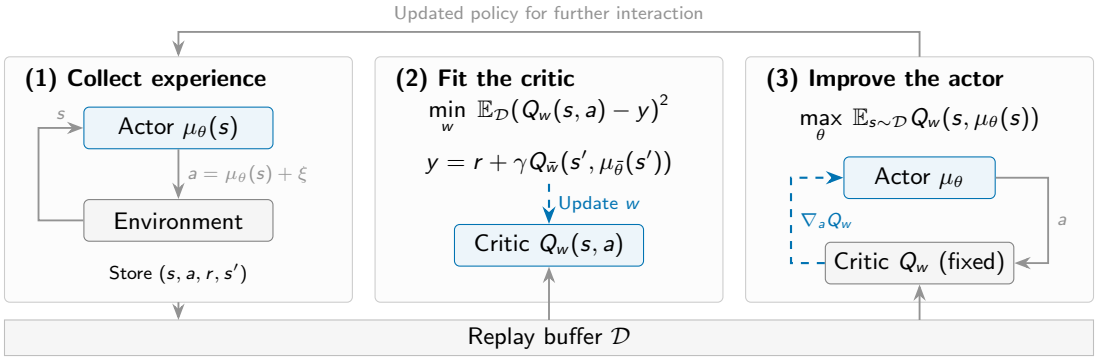
$$\mathcal{C} = \{s \in \mathbb{R}^n : h(s) \geq 0\}.$$

Keep the state in $\mathcal{C}$ throughout training, starting from $s_0 \in \mathcal{C}$.

The control input also satisfies $a_{\min} \leq a_t \leq a_{\max}$.

Policy Gradient-Based RL: DDPG

Deep Deterministic Policy Gradient

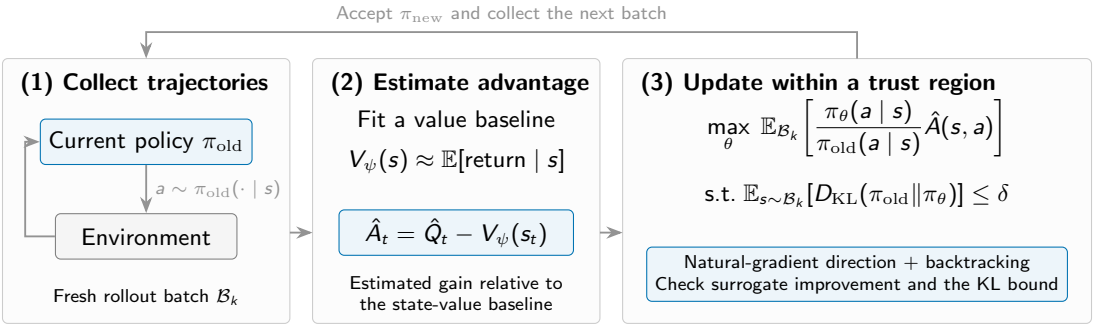


- **Off-policy:** reuse past transitions through minibatches from the replay buffer.
- **Deterministic actor:** follow the critic's action gradient. Add noise for exploration.
- **Target networks:** slowly updated $(\bar{\theta}, \bar{w})$ stabilize the critic's targets.

T. P. Lillicrap et al., "Continuous control with deep reinforcement learning," ICLR, 2016.

Policy Gradient-Based RL: TRPO

Trust Region Policy Optimization



- **On-policy:** collect a new batch with the current policy at each iteration.
- **Stochastic policy:** increase the probability of actions with positive advantage.
- **Trust region:** constrain the average KL divergence to limit policy changes.

J. Schulman et al., "Trust Region Policy Optimization," ICML, 2015.

Gaussian Processes

Estimate the model residual $d(s)$ from recorded transitions:

$$\hat{d}(s_t) = s_{t+1} - f(s_t) - g(s_t)a_t, \qquad a_t : \text{ applied control input.}$$

Fit a GP to each residual component, with a zero-mean prior, kernel k and Gaussian noise variance σ^2_{noise}.

Posterior prediction

$$\mu_d(s) = \mathbf{k}_s^\top (K + \sigma^2_{\text{noise}} I)^{-1} \mathbf{y},$$
$$\sigma_d^2(s) = k(s,s) - \mathbf{k}_s^\top (K + \sigma^2_{\text{noise}} I)^{-1} \mathbf{k}_s.$$

$K_{ij} = k(s_i, s_j), \quad \mathbf{k}_s = [k(s_i, s)]_{i=1}^N, \quad \mathbf{y} = [\hat{d}(s_i)]_{i=1}^N.$

GP confidence bounds: $\boxed{\mu_d(s) - k_\delta \sigma_d(s) \le d(s) \le \mu_d(s) + k_\delta \sigma_d(s).}$

- **Paper setting:** refit the GP between policy iterations using the latest $\approx 1{,}000$ transitions.
- **Computational cost:** exact GP inference requires $\mathcal{O}(N^3)$ matrix factorization. Sparse or local GP approximations reduce this cost.

Cheng et al., AAAI 2019; C. E. Rasmussen and C. K. I. Williams, Gaussian Processes for Machine Learning, 2006.

Control barrier functions

For $h(s) = p^\top s + q$ and $0 \le \eta \le 1$, the GP confidence bounds give the robust condition

$$\underbrace{p^\top [f(s_t) + g(s_t)a_t + \mu_d(s_t)] + q}_{\text{predicted next-step barrier value}} - \underbrace{k_\delta |p|^\top \sigma_d(s_t)}_{\text{uncertainty margin}} \ge (1-\eta) h(s_t). \tag{2}$$

The CBF controller minimizes control effort while allowing a penalized safety slack:

$$\begin{aligned} (a_t^\star, \epsilon_t^\star) = \arg\min_{a_t,\, \epsilon \ge 0} \ & \tfrac{1}{2}\|a_t\|_2^2 + K_\epsilon \epsilon \\ \text{s.t.} \quad & p^\top [f(s_t) + g(s_t)a_t + \mu_d(s_t)] + q - k_\delta |p|^\top \sigma_d(s_t) \ge (1-\eta) h(s_t) - \epsilon, \\ & a_{\text{low}} \le a_t \le a_{\text{high}}. \end{aligned} \tag{3}$$

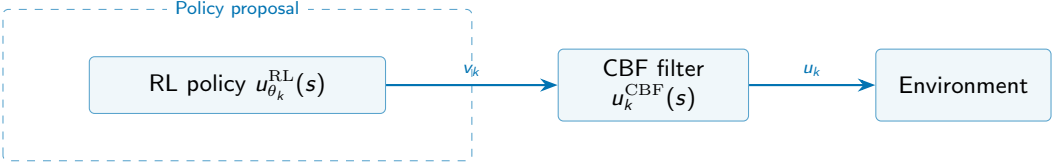
The controller applies $a_t^\star$. The safety slack $\epsilon \ge 0$ has penalty weight $K_\epsilon > 0$.

Safety guarantee

Assume the GP confidence bounds hold on the sets below and the required control inputs are feasible.

- If $\epsilon_t^\star = 0$ throughout $\mathcal{C}$, then $s_0 \in \mathcal{C} \Rightarrow s_t \in \mathcal{C}$ for all t.
- If $\eta > 0$ and $\epsilon_t^\star \le \bar{\epsilon}$ throughout $\mathcal{C}_{\bar{\epsilon}}$, then $\mathcal{C}_{\bar{\epsilon}} = \{s : h(s) \ge -\bar{\epsilon}/\eta\}$ is forward invariant.

CBF-Based Control with RL: Compensating



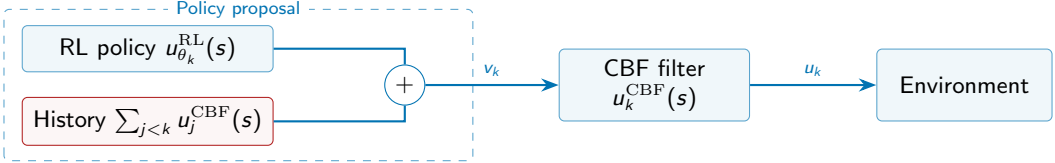
Proposal: $v_k = u_{\theta_k}^{\mathrm{RL}}$; applied input: $u_k = v_k + u_k^{\mathrm{CBF}}$.
The optimal correction $u_k^{\mathrm{CBF}}(s_t) = \Delta u_t^\star$ is obtained by solving the QP:

$$
\begin{aligned}
&(\Delta u_t^\star, \epsilon_t^\star) = \arg\min_{\Delta u_t,\, \epsilon \geq 0} \ \tfrac{1}{2}\|\Delta u_t\|_2^2 + K_\epsilon \epsilon \\
&\text{s.t.} \quad p^\top \left[f(s_t) + g(s_t)\big(v_k(s_t) + \Delta u_t\big) + \mu_d(s_t) \right] + q - k_\delta |p|^\top \sigma_d(s_t) \geq (1-\eta) h(s_t) - \epsilon, \\
&\qquad a_{\mathrm{low}} \leq v_k(s_t) + \Delta u_t \leq a_{\mathrm{high}}.
\end{aligned}
\tag{4}
$$

Policy update

RL optimizes $u_{\theta_k}^{\mathrm{RL}}$, while the deployed controller is u_k. The next proposal, $u_{\theta_{k+1}}^{\mathrm{RL}}$, does not explicitly retain past CBF corrections.

CBF-Based Control with RL: Guiding



Proposal: $v_k = u_{\theta_k}^{\mathrm{RL}} + \sum_{j<k} u_j^{\mathrm{CBF}}$; applied input: $u_k = v_k + u_k^{\mathrm{CBF}}$.
The optimal correction $u_k^{\mathrm{CBF}}(s_t) = \Delta u_t^\star$ is obtained by solving the QP:

$$
\begin{aligned}
&(\Delta u_t^\star, \epsilon_t^\star) = \arg\min_{\Delta u_t,\, \epsilon \geq 0} \ \tfrac{1}{2}\|\Delta u_t\|_2^2 + K_\epsilon \epsilon \\
&\text{s.t.} \quad p^\top \left[f(s_t) + g(s_t)\big(v_k(s_t) + \Delta u_t\big) + \mu_d(s_t) \right] + q - k_\delta |p|^\top \sigma_d(s_t) \geq (1-\eta) h(s_t) - \epsilon, \\
&\qquad a_{\mathrm{low}} \leq v_k(s_t) + \Delta u_t \leq a_{\mathrm{high}}.
\end{aligned}
\tag{5}
$$

Policy update

Retain the latest correction in the next proposal: $v_{k+1}(s) = u_{\theta_{k+1}}^{\mathrm{RL}}(s) + \sum_{j=0}^{k} u_j^{\mathrm{CBF}}(s)$. RL optimizes the proposal using the previously deployed controller as a reference. The current CBF filter enforces safety.

Efficient implementation

Exact evaluation of all previous corrections requires storing historical policies and solving an increasing number of QPs.

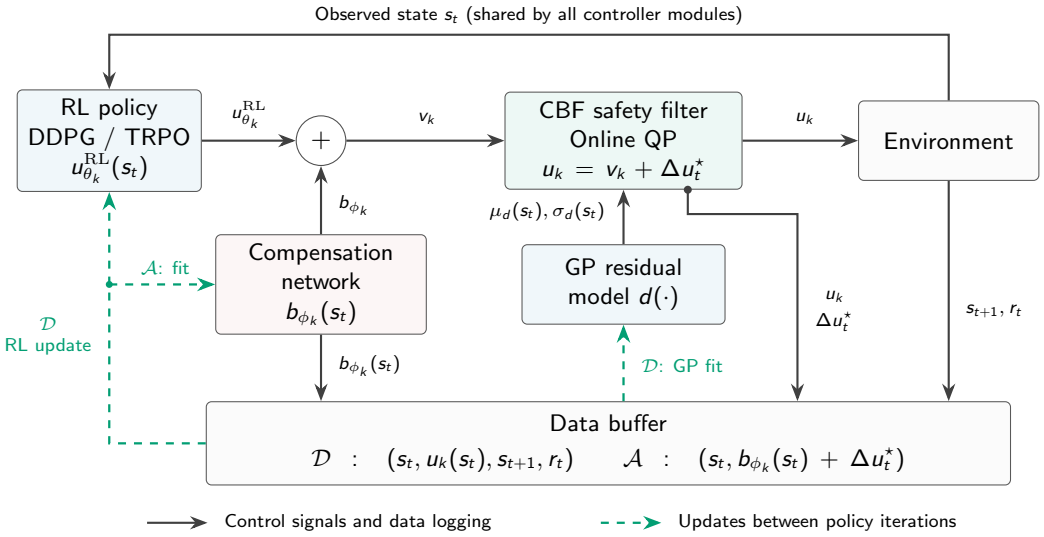
Compensation network

A feedforward network approximates the accumulated CBF corrections:

$$b_{\phi_k}(s) \approx \sum_{j=0}^{k-1} u_j^{\mathrm{CBF}}(s), \qquad u_k(s) = u_{\theta_k}^{\mathrm{RL}}(s) + b_{\phi_k}(s) + u_k^{\mathrm{CBF}}(s). \tag{6}$$

1. Fit $b_{\phi_{k+1}}$ to the targets $b_{\phi_k}(s_t) + \Delta u_t^\star$ collected in iteration k.
2. Online control requires two network evaluations and one QP per step, independent of the number of policy iterations.
3. The approximation changes the proposed input. The final CBF filter still enforces the safety constraints, but the approximation does not generally preserve the TRPO performance bound.

Control architecture



Reproduction overview

We corrected implementation errors and revised the modeling and experimental settings in the original code. The following results focus on the revised implementation, with original-code reruns for comparison. The reproduction report details the changes.

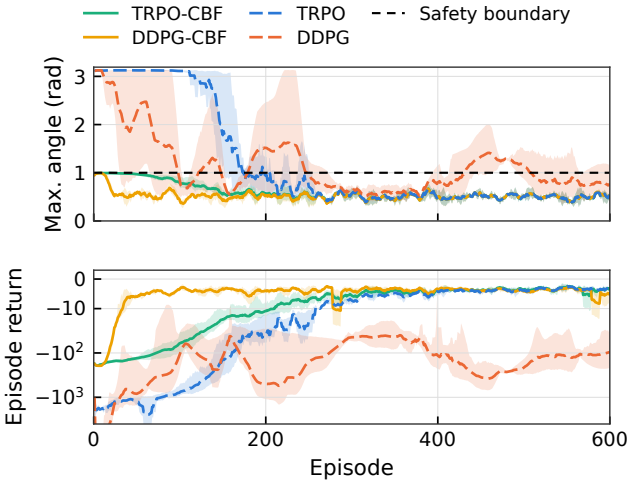
Task	Training per seed	Safety constraint
Inverted pendulum	600 episodes, 200 steps each	$\lvert\theta\rvert \leq 1$ rad
Car following (5 cars)	3×10^5 steps	Car 4: front/rear gaps ≥ 2 m

Four algorithms, seeds 1–3. Within each seed, algorithms share the environment and initial-state sequences. Car runs also share the acceleration-noise sequences.

Reruns of the original code

- **Pendulum:** in the first 600 episodes, TRPO-CBF recorded 1, 9 and 332 violating episodes across seeds 1–3. DDPG-CBF recorded 37–127 and continued to rely on CBF corrections.
- **Car following:** both CBF variants avoided collisions, but the gap fell below 2 m in 12 episodes. DDPG-CBF did not converge.

Inverted pendulum: training results



Safety

TRPO-CBF: **0/1800** violating episodes.
DDPG-CBF: **1/1800**, in seed 3, episode 7, with positive slack.

Learning speed

Episodes to the first 20-episode mean return above -10.

DDPG-CBF	36–45
TRPO-CBF	157–244
TRPO	264–268
DDPG	Not reached

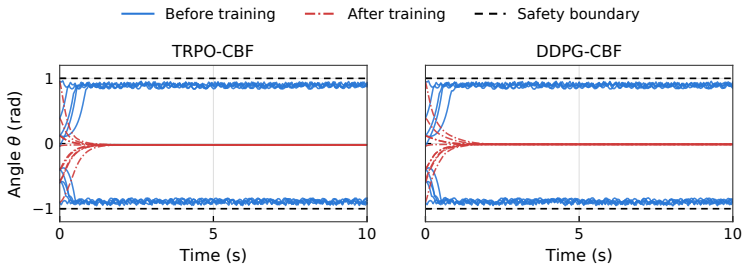
Comparable final returns

TRPO: -3.33 TRPO-CBF: -3.52.
Mean over the last 100 episodes.

Revised code, seeds 1–3. Curves use 10-episode smoothing. Lines show seed means, shading the min–max range. Return axis: symmetric log scale.

Inverted pendulum: closed-loop evaluation

We compare the untrained and final policies from the same ten initial states (seed 1), using deterministic actions for 10 s per rollout. Each stage uses its corresponding compensation network and GP, with no GP updates during evaluation.



Before training

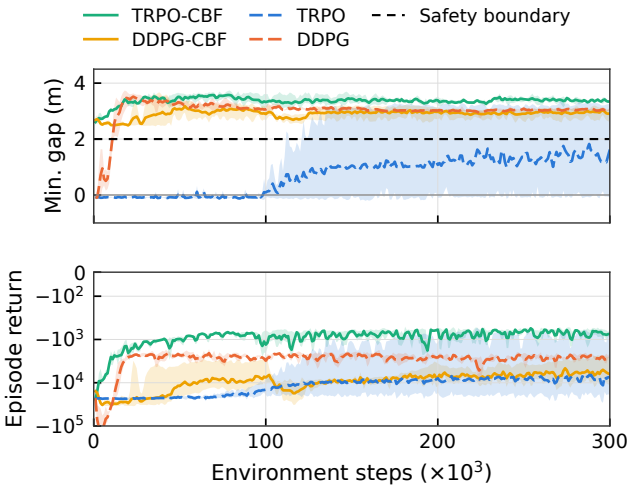
CBF corrections keep the pendulum near $|\theta| \approx 0.9$ rad.

After training

Upright within 1–2 s, with $u^{\mathrm{CBF}} = 0$ throughout evaluation.

Training correction RMS: 3.3 to 0.02 N · m (TRPO-CBF), 3.7 to 0.01 N · m (DDPG-CBF).

Car following: training results



Safety

Both CBF variants: no collisions or 2 m gap violations.
Minimum gap: **2.33 m**.

Learning speed

First 50-episode mean return above -5000:
TRPO-CBF: **11,000–14,000** steps.
DDPG: 20,000–23,000 steps.
Final TRPO-CBF return: -740.

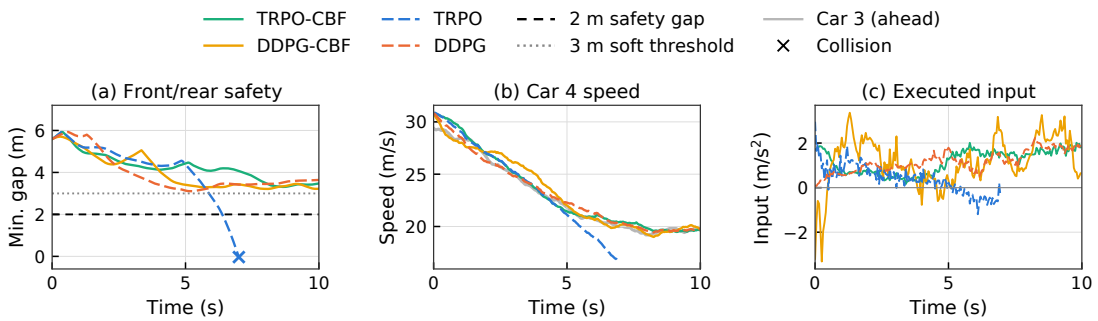
DDPG-CBF

Lower final return than DDPG: -5856 vs. -2586.
Compensation RMS: 92–98 m/s^2, near the ± 100 limit.

Revised code, seeds 1–3. Curves: 10-episode smoothing, seed mean and min–max range. Final return: last 100 episodes. Return axis: symmetric log scale.

Car following: closed-loop evaluation

We compare the final policies as the front car slows down. Each 10 s simulation uses deterministic actions, a common initial state and noise sequence, and fixed GP models. The plots show seed 1.



Seed 1

Minimum gaps are 3.1–3.3 m for both CBF variants and DDPG. TRPO is rear-ended at 7.0 s.

Across seeds

TRPO-CBF needs no CBF correction in any seed. DDPG-CBF needs it in 62% of steps (seed 2) and 55% (seed 3).

Conclusions

1. **Training safety.** Both car-following CBF variants remained collision-free. Pendulum TRPO-CBF had no violations, while DDPG-CBF had one early violation.
2. **Learning speed and final return.** Both pendulum CBF variants and car-following TRPO-CBF reached return thresholds earlier. Final returns were not consistently higher than the baselines.
3. **CBF corrections and compensation.** CBF corrections became negligible for both pendulum variants and car-following TRPO-CBF. Car-following DDPG-CBF showed compensation saturation and underperformed DDPG.

Comparison scope

Car dynamics, reward and episode length changed, so returns cannot be compared directly with the paper. The experiments do not isolate individual changes through ablations.

Sources: reproduction report, Sections 3–4 and Supplement C. These findings refer to the revised implementation.

Research plan: safety-guaranteed reinforcement learning

Goal: develop RL methods with safety and convergence guarantees for robotic systems.

1. **Theory: policy optimization and convergence**

   Study how CBF safety filtering affects policy updates and convergence. Develop learning rules that improve control performance while satisfying safety constraints.

2. **Theory: safety guarantees under model uncertainty**

   Study how limited data and model errors affect CBF safety guarantees. Establish conditions for maintaining feasibility when several safety constraints must be satisfied at the same time.

3. **Application: safe humanoid locomotion**

   Evaluate policies trained with CBFs on humanoids under changing terrain, payloads and sensing errors. Develop policy verification methods and lightweight online safeguards to improve deployment safety while meeting real-time constraints.